3. How Machines Learn, Part 1

Linear regression and the cost function

Week 3 · CE/ARCE 2XX AI Literacy for Civil and Architectural Engineering



COLLEGE OF ENGINEERING
Civil & Architectural Engineering & Mechanics

Agenda

Today's plan

1. Quiz on last week (5–8 min)
2. The problem: predict concrete strength from the water–cement ratio
3. The simplest model: a straight line, and its notation
4. Measuring error: the cost function J(w, b)
5. The cost landscape: a bowl, and its bottom
6. How good is the fit? MAE and R^2 — and the lab

Recap

Last week in 60 seconds

1 Rules → learning

Machine learning takes **data + answers** and produces the rules: a model, $\hat{y} = f(x)$.

2 X and y

Features X go in; the target y comes out. One row = one example.

3 Three ways to learn

Supervised (regression, classification), unsupervised, reinforcement.

4 Data is a design material

Quality, quantity, labels, representativeness.

Learning objectives

By the end of today, you can…

- ✓ Write a linear model $\hat{y} = w \cdot x + b$ and explain w and b in engineering units
- ✓ Use the notation $x^{(i)}$, $y^{(i)}$, m, and $\hat{y}$ to describe a dataset
- ✓ Compute the squared-error cost J(w, b) by hand for a few points
- ✓ Read a cost curve and a contour map, and say where the best (w, b) is
- ✓ Judge a fit with MAE (in MPa) and R^2, and name the limits of a straight line

01

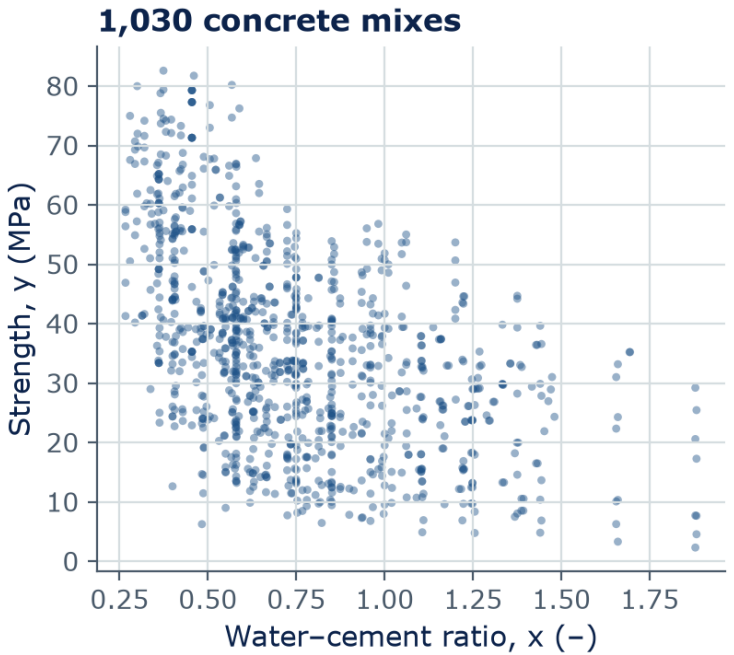
The simplest model

A straight line through concrete data

The problem

Can we predict strength from the water–cement ratio?

- **Water–cement ratio** (w/c) = water ÷ cement, both in kg/m³
- Engineers have known for a century: **more water per kg of cement → weaker concrete**
- Each dot = one mix from our 1,030-mix dataset
- Goal: a model that turns **x = w/c** into a predicted strength **ŷ** in MPa



Data: Yeh (1998), UCI concrete dataset (course data folder)

Notation: one row of data = one example ($x^{(i)}$, $y^{(i)}$)

i	Water (kg/m³)	Cement (kg/m³)	$x^{(i)}$ = w/c	$y^{(i)}$ (MPa)
1	162	540	0.300	79.99
2	162	540	0.300	61.89
3	228	332.5	0.686	40.27
4	228	332.5	0.686	41.05
…	…	…	…	…

m = 1,030

x	input (feature): here the w/c ratio
y	target: measured strength, MPa
$\hat{y}$	prediction (“y-hat”), MPa
m	number of examples (1,030)
$x^{(i)}$, $y^{(i)}$	the i-th example — (i) is an index, not a power

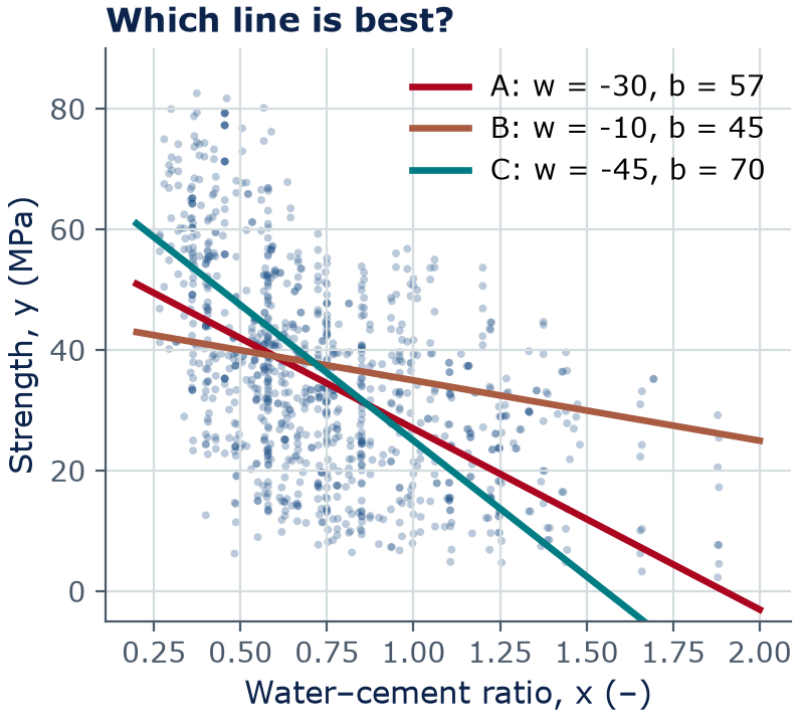
The model is a straight line with two parameters, w and b

$$\hat{y} = f_{w,b}(x) = w \cdot x + b$$

x	input: water–cement ratio (no units)
ŷ	predicted compressive strength (MPa)
w	weight = slope: change in ŷ per 1.0 change in x (MPa)
b	bias = intercept: ŷ when x = 0 (MPa)
w, b	the parameters — the numbers the machine learns

Different w and b give different lines — which is best?

- Each pair (w, b) is a different line — a different model
- Change w → the line tilts. Change b → the line moves up or down.
- By eye, **A** looks best: B is too flat, C is too steep
- Engineers need more than "by eye" — we need a **number** that scores each line



Quick calculation: use a model

2 min

Solo · then compare

- Use line **A**: $\hat{y} = -30 \cdot x + 57$ (MPa)
- Predict the strength of a mix with **x = 0.45**
- Then a mix with **x = 0.70**. Which is stronger, and by how much?

Check: $\hat{y}(0.45) = -13.5 + 57 = 43.5$ MPa · $\hat{y}(0.70) = -21 + 57 = 36$ MPa · difference 7.5 MPa

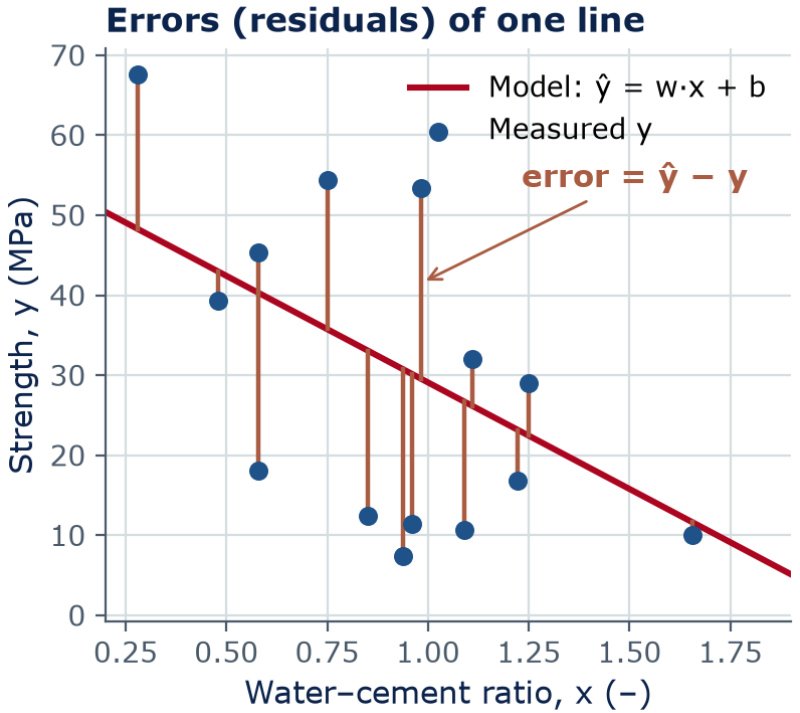
02

Measuring error

The cost function J(w, b)

For every example, the error is prediction minus truth

- **Error** (residual) of example i: $\hat{y}^{(i)} - y^{(i)}$, in MPa
- Positive → the model over-predicts (unsafe side for strength!)
- Negative → the model under-predicts
- A good line keeps all the vertical gaps small — but there are 1,030 of them



Measuring error

Why we square the errors

1 No canceling

Squares are never negative, so errors of +5 and −5 MPa cannot cancel out.

2 Big misses count more

A 10 MPa miss costs 100; two 5 MPa misses cost 50. Large errors are punished.

3 A smooth bowl

Squared error makes a smooth cost curve with one bottom — easy to find with calculus (next week).

The cost function: the average squared error, halved

$$J(w,b) = \frac{1}{2m}\sum_{i=1}^{m}\left(f_{w,b}(x^{(i)}) - y^{(i)}\right)^2$$

$f(x^{(i)})$	the prediction $\hat{y}^{(i)}$ for example i
$- y^{(i)}$	minus the measured value → the error
$(\ )^2$	squared, so every error counts as positive
Σ	add up over all m examples
$1/(2m)$	average; the ½ makes next week's math simpler

Your turn: compute J by hand for two lines

4 min

Pairs

Mix	x = w/c	y (MPa)
A	0.4	50
B	0.6	38
C	0.8	30

Line 1: $\hat{y} = -50 \cdot x + 70$

Line 2: $\hat{y} = -30 \cdot x + 60$

- For each line and each mix: compute **$\hat{y}$**
- Error = **$\hat{y} - y$**; then square it
- Add the three squares
- Divide by **$2m = 6$** → J
- Which line is better?

Answer: line 1 has the lower cost, so it fits better

Mix	x	y (MPa)	ŷ, line 1	$(\hat{y} - y)^2$	ŷ, line 2	$(\hat{y} - y)^2$
A	0.4	50	50	0	48	4
B	0.6	38	40	4	42	16
C	0.8	30	30	0	36	36
Sum				4		56
J = sum / 6				**0.67 MPa²**		9.33 MPa²

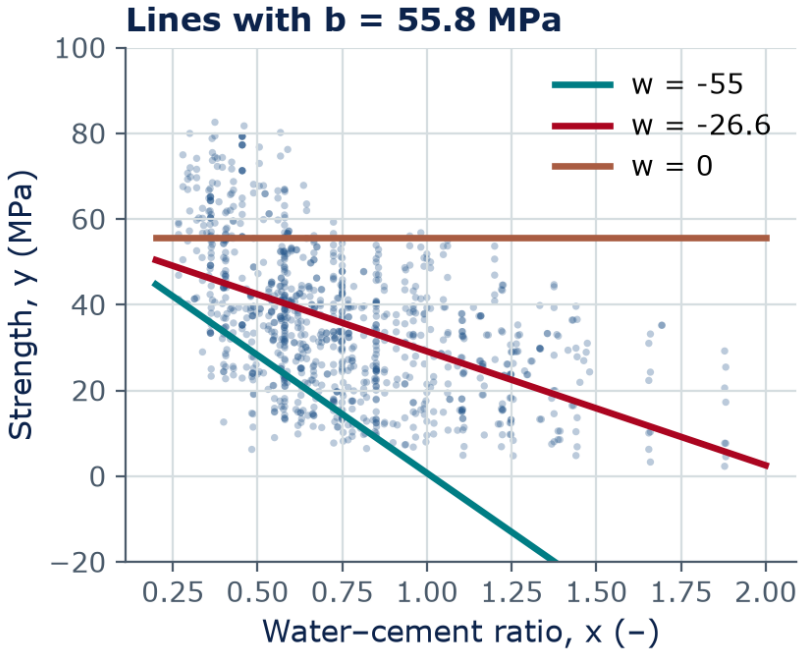
03

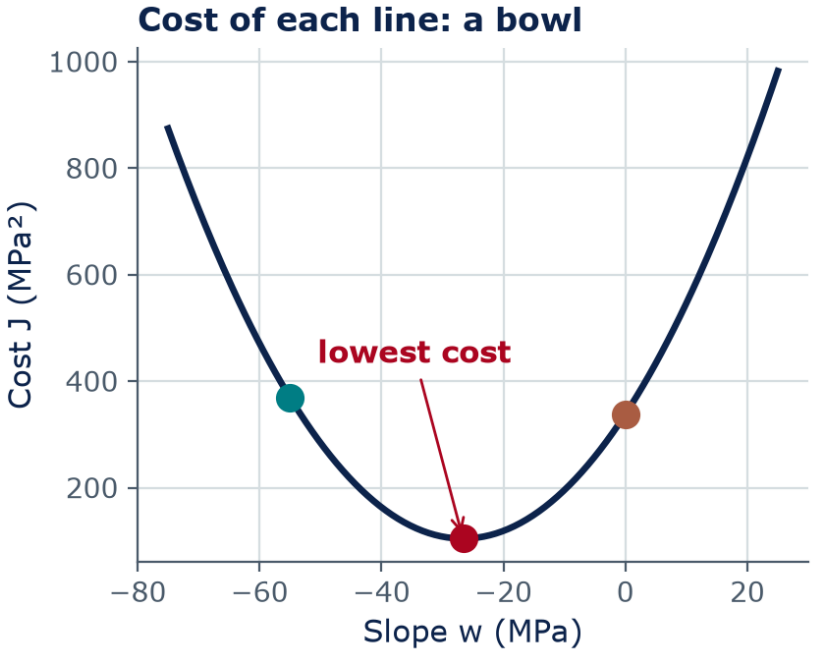
The cost landscape

Every line is a point on a bowl

The cost landscape

Each line has a cost; plotting the cost against w gives a bowl

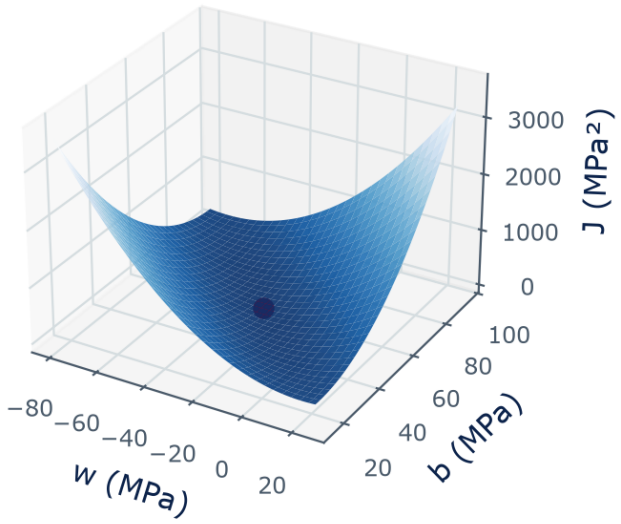




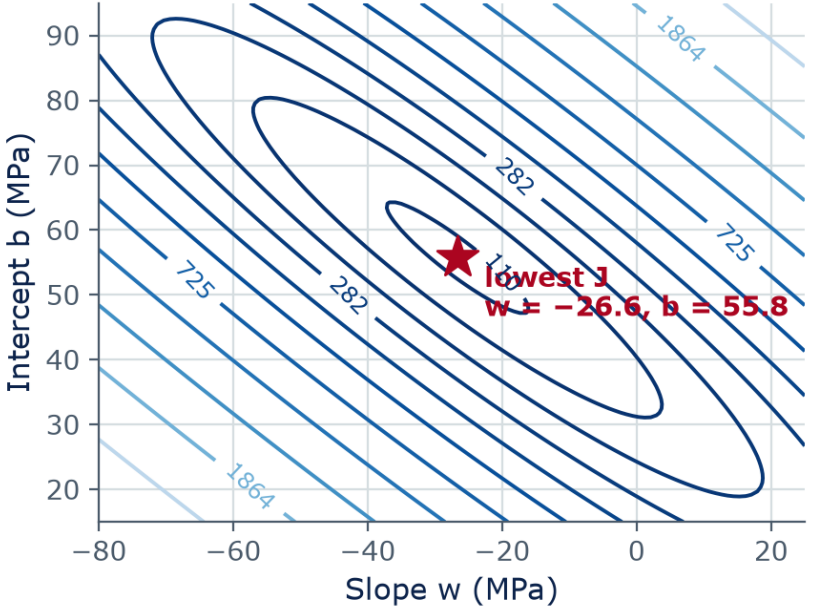
Left: three lines with the same b. Right: J for every w — the red line (w = -26.6 MPa) sits at the bottom.

With two parameters, the cost is a bowl over the (w, b) plane

Cost J(w, b): a 3-D bowl



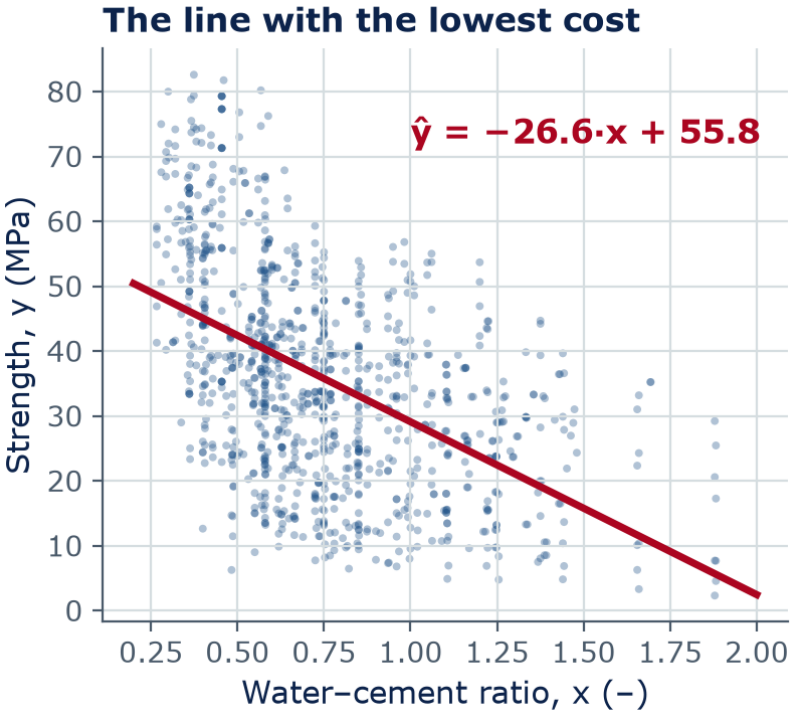
Same bowl from above: contour map



A contour map is like a topographic map: each ring joins (w, b) pairs with the same cost. The red star is the bottom.

The bottom of the bowl is the best-fit line

- Bottom of the bowl: **w ≈ −26.6 MPa**, **b ≈ 55.8 MPa** (J ≈ 104 MPa²)
- Each +0.1 in w/c → about **2.7 MPa** less strength
- At w/c = 0.5: $\hat{y} = -26.6 \times 0.5 + 55.8 \approx$ **42 MPa**
- Found here by least squares (`numpy.polyfit`); next week by **gradient descent**



Concept check

What happens to J?

Suppose every error of a model doubled (each $\hat{y}^{(i)} - y^{(i)}$ became twice as large). What happens to the cost J?

A J doubles

B J becomes 4 times larger

C J stays the same

D J is cut in half

Concept check

What happens to J? — answer

Suppose every error of a model doubled (each $\hat{y}^{(i)} - y^{(i)}$ became twice as large). What happens to the cost J?

A J doubles

B J becomes 4 times larger

C J stays the same

D J is cut in half

J averages squared errors: $(2e)^2 = 4e^2$, so J becomes 4 times larger.

04

How good is the fit?

Metrics in engineering units — and the limits of a line

How good is the fit?

Two metrics engineers can read: MAE and R^2

$$\mathrm{MAE} = \frac{1}{m}\sum_{i=1}^{m}\left|\hat{y}^{(i)} - y^{(i)}\right|$$

MAE: mean absolute error

- The average size of a miss, in MPa
- Smaller is better; 0 = perfect

$$R^2 = 1 - \frac{\sum_i (y^{(i)} - \hat{y}^{(i)})^2}{\sum_i (y^{(i)} - \bar{y})^2}$$

R^2: coefficient of determination

- Share of the variation in y the model explains
- 1 = perfect · 0 = no better than the average $\bar{y}$

How good is the fit?

Our best line (all data): right trend, but large errors

11.9 MPa

MAE (all data): the average miss over all 1,030 mixes

0.25

R^2 (all data): the line explains about 25% of the variation in strength

2 of 8

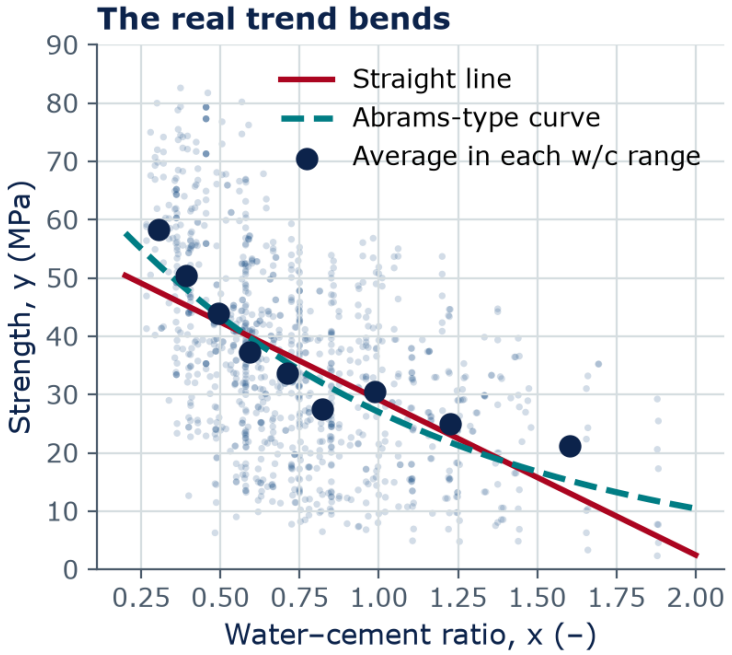
Inputs used: w/c uses only water and cement — age, slag, fly ash… are ignored

Typical specified concrete strengths are roughly 20–40 MPa, so a 12 MPa average miss is too large to design with.

How good is the fit?

The real relationship is curved — a line is an approximation

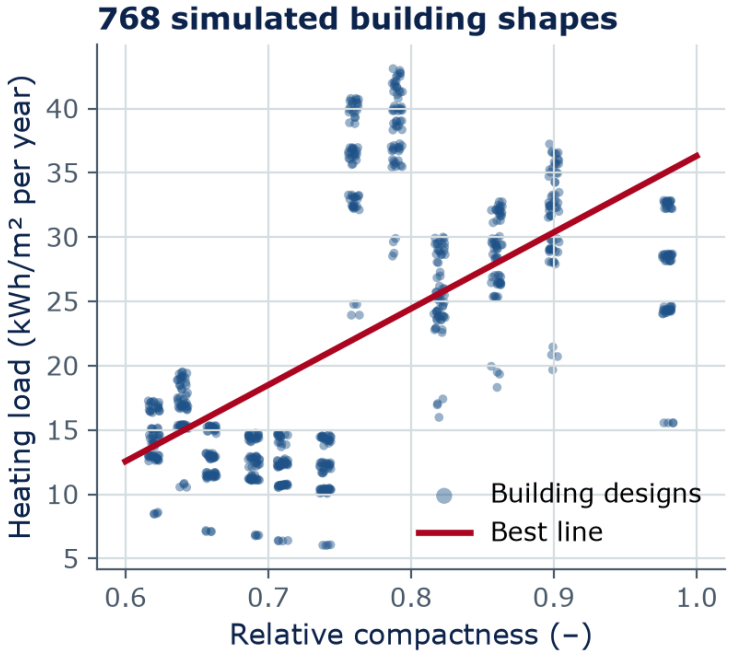
- **Abrams' law** (1918): strength = A / B^(w/c), with A and B fitted to the materials
- Averages (navy dots) fall steeply at low w/c and flatten at high w/c
- The line under-predicts at both ends and over-predicts in the middle
- Fixes: more features (Week 4) and **more flexible models** (neural networks, Week 5)



Curve: A = 70 MPa, B = 2.58 (R^2 0.28 vs 0.25)

Same recipe on building data: look before you trust a line

- x = relative compactness (–), y = annual heating load (kWh/m²)
- Best line: $\hat{y} = 59.4 \cdot x - 23.1$
- $R^2 = 0.39$, MAE = 6.1 kWh/m²
- But look: the points form **two separate groups** — one line hides that
- Why two groups? **HW1 Part C6** asks you to find out. Always plot first.



Data: Tsanas & Xifara (2012), UCI energy-efficiency dataset

The recipe

The machine-learning recipe, so far

1 · Model

- Choose a form:
- $\hat{y} = w \cdot x + b$

2 · Cost

- Score it with
- $J(w, b)$ — average squared error

3 · Optimize

- Find the w, b with the lowest J
- (next week: gradient descent)

4 · Evaluate

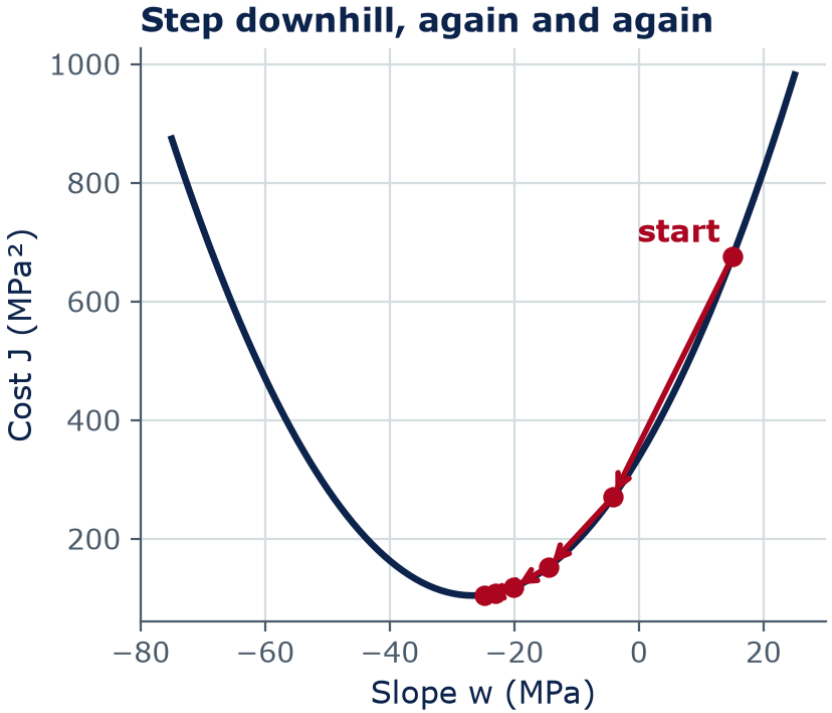
- MAE, R^2, plots
- Would an engineer trust it?

The same four steps train neural networks and large language models — only the model gets bigger

Next week

Preview: find the bottom by walking downhill

- Bigger models have no formula for the bottom of the bowl
- Instead: start anywhere, check the **slope**, take a step downhill
- Repeat until the cost stops falling
- That is **gradient descent** — how almost every modern AI model is trained



Lab 03

Lab day: model representation and cost

LAB

03

Bring a charged laptop

Open the notebook from the course GitHub page

- Plot strength vs w/c; write `predict(x, w, b)` and watch the line move as you change w and b
- Write `compute_cost(x, y, w, b)` — today's J, in code
- Plot the cost bowl and contour map; search for the best (w, b); check with `numpy.polyfit`
- **Exit ticket:** 3 short questions you answer alone, without AI, at the end of lab

Wrap-up

Key takeaways

1. The simplest model is a line: $\hat{y} = w \cdot x + b$; learning = choosing w and b
2. Error = $\hat{y} - y$. The cost $J(w, b)$ = average squared error ÷ 2 — one number per line
3. Plotting J over w and b gives a bowl; the **bottom** is the best-fit line
4. Our concrete line (all data): MAE ≈ 12 MPa, $R^2 \approx 0.25$ — right trend, big errors
5. The real trend is curved (Abrams' law): a line is an approximation — always plot your data

Learn more

Readings and resources

- **Google ML Crash Course: Linear regression** — the Linear regression and Loss pages (whole module about 80 min)
- **Elements of AI, Chapter 4, Section III: Regression** — short reading with exercises, about 15 min
- **Course GitHub: Week 3 README** — key terms and check-yourself questions
- **Abrams, D. A. (1918), Design of Concrete Mixtures** — Lewis Institute Bulletin 1 — the origin of the w/c law (for the curious)